CSE6730 Project Description

Group 31: Jiasheng Cao, Ruixin Liu, Zijun Meng, Yuming Sun

1 Abstract

This project aims to develop an integrated mechanistic simulation framework that couples biological wastewater treatment with hydroponic crop production. While conventional models typically isolate municipal nutrient removal from agricultural nutrient uptake, our system-level approach bridges this gap by linking the Activated Sludge Model No. 1 (ASM1) with Michaelis-Menten and Monod-type plant growth kinetics. The simulation is developed in Python, utilizing a fourth-order Runge-Kutta (RK4) numerical integration scheme to resolve the coupled nonlinear ordinary differential equations governing microbial growth, nitrogen transformation, and continuous-flow mass balances. Current progress focuses on the upstream nitrification module, modeling the two-step oxidation of ammonia to nitrate by autotrophic bacteria within a Continuous Stirred-Tank Reactor (CSTR). Preliminary simulations successfully capture the dynamic competition between microbial biomass accumulation and hydraulic dilution, identifying critical washout conditions that dictate system start-up requirements. Future development will fully integrate the downstream hydroponic module, enabling predictive analysis and parameter optimization for stable nutrient recovery across the wastewater-agriculture interface.

2 System Description

Wastewater Treatment System Description: The system modeled in this project is a simplified aerobic nitrification reactor configured as a CSTR. The purpose of this unit is to convert ammonium in wastewater into oxidized nitrogen species before the treated effluent is supplied to a downstream vegetable growth model. In this coupled framework, the wastewater treatment unit functions as the upstream nitrogen transformation step, while the crop model uses the simulated effluent composition as an input for plant growth analysis.

The reactor is represented as a single completely mixed tank operated under continuous-flow conditions. Wastewater enters the reactor at a constant flow rate and leaves at the same flow rate, thus reactor volume is constant during simulation. Within this reactor, nitrification is represented as a two-step biological process: ammonium is first oxidized to nitrite by ammonia-oxidizing bacteria (AOB), and nitrite is then oxidized to nitrate by nitrite-oxidizing bacteria (NOB). This configuration was selected because it captures the main structure of nitrification that is suitable for producing the nitrogen species required by the downstream crops.

Several assumptions were adopted to simplify the model. First, the reactor was assumed to be perfectly mixed, meaning that substrate and biomass concentrations are spatially uniform throughout the tank. Second, only aerobic nitrification was considered, other nitrogen conversion pathways were not included. Third, dissolved oxygen was treated as a constant operating condition rather than a dynamic state variable. Fourth, pH and temperature effects were considered in this system, whereas alkalinity limitation, and inhibition by free ammonia or free nitrous acid were neglected. Fifth, biomass loss was represented through endogenous decay and hydraulic washout, consistent with the idealized CSTR framework used in this project.

Overall, this simplified system configuration provides a reasonable mechanistic representation of an aerobic nitrification unit and is suitable for evaluating how influent ammonium is transformed into nitrate for subsequent reuse in vegetable production.

Hydroponic Plant Growth Simulation Module Description: The plant growth simulation module represents the downstream crop-production component of the integrated wastewater–hydroponic framework. Its purpose is to translate nutrient conditions supplied at the wastewater–agriculture interface into time-resolved plant biomass accumulation. In the overall system, the upstream wastewater-treatment module is intended to generate reclaimed nutrients after treatment, and these will then serve as source of fertilizers and input nutrient variables to the downstream cultivation module. In this way, the cultivation module provides the link between nutrient recovery and plant production performance across the integrated system.

Mechanistic simulation of hydroponic plant systems has commonly focused on biomass dynamics together with nutrient uptake and accumulation, particularly under nutrient-limited conditions.[11,12,15] Such studies typically use ordinary differential equation (ODE) frameworks in which biomass growth is governed by nutrient availability and related environmental forcing. Building on this literature, the present hydroponic plant growth module is formulated as a general mechanistic crop-growth simulator based on specific growth-rate formulations and a biomass governing equation. To represent alternative ways of describing nutrient-limited plant growth, the current implementation includes five candidate formulations for the specific growth rate, μ: a nitrogen-focused Dynamic-μ model, a category-aware RMND-μ model, a shared SMND-μ model, a Liebig minimum Monod model, and a multiplicative Monod model. The Monod-type nutrient limitation provides a standard starting point for hydroponic growth simulation, while Dynamic-μ formulations offer a compact extension for representing temporally varying relative growth behavior under nutrient-limited conditions.[17,18,19]

Integrated System Description: In this project, the wastewater treatment module and the crop cultivation module are designed as an integrated system. The two subsystems are connected through nitrogen flow. The wastewater treatment module serves as the upstream component. It transforms influent ammonium into nitrate through biological nitrification. The output of this module is the effluent nitrogen composition, including ammonium, nitrite, and nitrate concentrations. The crop cultivation module serves as the downstream component. It takes the nitrogen concentrations from the wastewater model as input and uses them to simulate plant nutrient uptake and biomass growth. Through this connection, the system links wastewater treatment with agricultural production. Instead of treating wastewater only for discharge, the model allows us to study how treated nitrogen can be reused for crop growth. At the current stage, the two modules are developed and tested separately. In the next stage, they will be fully coupled to enable system-level simulation and analysis.

3 Conceptual Model and Mathematical Framework

Mechanistic Modeling of Nitrogen Transformation: The nitrogen transformation process in the reactor was described using a mechanistic model that combines kinetic expressions for microbial growth and substrate utilization with mass-balance equations for soluble nitrogen species and biomass concentrations. In this framework, the kinetics define how AOB and NOB respond to substrate and environmental conditions, while the mass balances describe how ammonium, nitrite, nitrate, and microbial biomass change over time in the CSTR.

The growth of AOB and NOB was represented using Monod-type kinetic expressions. For AOB, the specific growth rate was defined as:

$$\mu_{AOB} = \mu_{AOB,max} \left(\frac{S_{NH4}}{K_{NH4} + S_{NH4}} \right) \left(\frac{S_{O2}}{K_{O2,AOB} + S_{O2}} \right) X_{AOB} \tag{1}$$

This equation reflects that AOB growth depends on both ammonium availability and dissolved oxygen concentration. Likewise, the specific growth rate of NOB was defined as:

$$\mu_{NOB} = \mu_{NOB,max} \left(\frac{S_{NO2}}{K_{NO2} + S_{NO2}} \right) \left(\frac{S_{O2}}{K_{O2,NOB} + S_{O2}} \right) X_{NOB} \tag{2}$$

which represents the second nitrification step, where nitrite is oxidized to nitrate under aerobic conditions. In both cases, the Monod form accounts for substrate limitation by allowing the microbial growth rate to increase with substrate concentration and gradually approach a maximum value.

To account for environmental effects, the maximum specific growth rates were further corrected for temperature and pH. Temperature dependence was described using an Arrhenius-type relationship:

$$\mu_{\max,T} = \mu_{\max,20} \cdot \theta^{(T-20)} \tag{3}$$

and pH dependence was incorporated as:

$$\mu_{\max,pH} = \mu_{\max} \cdot (1 - 0.833 \cdot (7.2 - pH)) \tag{4}$$

These correction terms allow the model to reflect the influence of operating conditions on nitrifying activity in a simplified way.

The microbial growth rates were then linked to nitrogen conversion through biomass yield coefficients. The ammonium utilization rate by AOB and the nitrite utilization rate by NOB were expressed as:

$$r_{AOB} = \frac{\mu_{AOB} X_{AOB}}{Y_{AOB}} \tag{5}$$

$$r_{NOB} = \frac{\mu_{NOB} X_{NOB}}{Y_{NOB}} \tag{6}$$

These relationships connect microbial growth with substrate removal, thereby linking biomass dynamics directly to nitrogen transformation.

Based on these kinetic expressions, the system was described using a set of five coupled ODEs, representing the dynamic mass balances of the three soluble nitrogen species and the two key functional microbial groups. The nitrogen balances in the CSTR were written as:

$$\frac{dS_{NH4}}{dt} = \frac{Q}{V}(S_{NH4,in} - S_{NH4}) - r_{AOB} \tag{7}$$

$$\frac{dS_{NO2}}{dt} = \frac{Q}{V}(S_{NO2,in} - S_{NO2}) + r_{AOB} - r_{NOB} \tag{8}$$

$$\frac{dS_{NO3}}{dt} = \frac{Q}{V}(S_{NO3,in} - S_{NO3}) + r_{NOB} \tag{9}$$

These equations represent, respectively, ammonium consumption by AOB, transient nitrite production and consumption, and nitrate accumulation as the final nitrification product.

The biomass balances for AOB and NOB were expressed as:

$$\frac{dX_{AOB}}{dt} = -\frac{Q}{V}X_{AOB} + (\mu_{AOB} - b_{AOB})X_{AOB} \tag{10}$$

$$\frac{dX_{NOB}}{dt} = -\frac{Q}{V}X_{NOB} + (\mu_{NOB} - b_{NOB})X_{NOB} \tag{11}$$

These equations account for biomass growth, endogenous decay, and hydraulic washout. Therefore, the simulated biomass concentrations reflect the balance between microbial growth and biomass loss under the specified reactor conditions.

Together, these equations provide a dynamic mechanistic representation of two-step nitrification in a completely mixed reactor and allow simulation of both nitrogen species and nitrifier biomass over time. The coupled ordinary differential equations were numerically integrated in Python using a hand-coded fourth-order RK4 scheme. At each time step, four intermediate slopes were calculated from the system dynamics, and these were combined to update the state variables, including ammonium, nitrite, nitrate, AOB biomass, and NOB biomass.

Mechanistic Modeling of Plant Growth: The plant growth module is based on a biomass-growth state equation in which nutrient availability determines the instantaneous specific growth rate, and the specific growth rate in turn governs biomass accumulation. Dry mass (DM) is used to represent biomass accumulation in plants. The biomass balance is written as

$$\frac{dDM}{dt} = \mu(t)\, DM \tag{12}$$

where $DM(t)$ denotes dry mass at time t, and $\mu(t)$ denotes the specific growth rate. This formulation represents biomass accumulation as an ordinary differential equation driven by the

instantaneous growth rate and is consistent with mechanistic hydroponic cultivation models in which biomass evolution is governed by nutrient availability [11,12,15].

Nutrient limitation is represented using Monod-type saturation functions,

$$f_i(S_i) = \frac{S_i}{K_{s,i} + S_i} \tag{13}$$

where S_i is the available concentration of nutrient i, and $K_{s,i}$ is the half-saturation coefficient for nutrient i. This project focused on nitrogen, phosphorus, and potassium concentrations (N, P, and K), which are macro-nutrients for plant growth. These nutrient are supplied by the upstream wastewater-treatment module in the integrated framework.

The Liebig minimum Monod model is written as

$$\mu = \mu_{\max} \min\left(f_N(N), f_P(P), f_K(K)\right) \tag{14}$$

which assumes that growth is controlled by the single most limiting nutrient. The multiplicative Monod model is written as

$$\mu = \mu_{\max}\, f_N(N)\, f_P(P)\, f_K(K) \tag{15}$$

which assumes simultaneous co-limitation by all included nutrients.

The Dynamic-μ model is written as

$$\mu = \mu_{\max} \frac{N}{K_s + N} + m \max(0, t - \tau) \tag{16}$$

where $\mu_{\max}$ is the maximum specific growth-rate coefficient, K_s is the nitrogen half-saturation coefficient, m is the slope of maturity-associated decline after the delay point, and τ is the delay time. This formulation extends the Monod-type nitrogen-limitation term by allowing the specific growth rate to decrease after a certain developmental stage. The RMND-μ model is a category-aware multi-nutrient formulation:

$$\mu = \mu_{\max,c}\, f_N(N)\, f_P(P)\, f_K(K) + m_c \max(0, t - \tau_c) \tag{17}$$

where subscript c indicates category-dependent parameters. The SMND-μ model uses a shared parameter set across retained conditions:

$$\mu = \mu_{\max}\, f_N(N)\, f_P(P)\, f_K(K) + m \max(0, t - \tau) \tag{18}$$

The Dynamic-μ formulation used here is based on published lettuce kinetic studies[18,19].

For each growth formulation, model parameters can be treated as cultivation-specific calibration variables, generically denoted by θ (e.g., including growth, half-saturation, and maturity-related terms as applicable to the selected model), rather than universal constants, which can be estimated by minimizing the mismatch between simulated dry-mass trajectories and measurements at the sampled growth days under the prescribed nutrient inputs for each treatment:

$$J(\theta) = \sum_j \left(DM_j^{obs} - DM_j^{pred}(\theta)\right)^2, \tag{19}$$

with the calibrated parameter set obtained as

$$\theta^* = \arg\min_{\theta} J(\theta). \tag{20}$$

4 Development Platform and Simulation Methodology

The simulation framework is developed in Python 3.x, using the NumPy library for numerical operations and Matplotlib for visualization of nitrogen and biomass dynamics. To maintain full control over the mathematical transparency of the system, the dynamic behaviors in both the upstream wastewater-treatment module and the downstream plant cultivation module are implemented using the fourth-order Runge–Kutta (RK4) method for numerical integration, rather than relying on external black-box ODE solver libraries. Collaborative development, version control, and file management are systematically maintained using GitHub. The complete repository can be accessed at: `https://github.com/Jiasheng-Cao/CSE6730_Final`.

5 Literature Review

Conventional wastewater treatment systems remove nutrients such as nitrogen and phosphorus to meet discharge standards before releasing effluent into natural water bodies. However, these nutrients could be reused for hydroponic crop production. Because wastewater models focus on nutrient removal and hydroponic studies treat nutrient solutions as fixed inputs, integrated mechanistic models linking treatment processes with hydroponic nutrient uptake and plant growth remain limited. Studies of integrated aquaponic simulation (i.e., fish-plant systems) provide examples of end-to-end module integration by explicitly representing nutrient generation and transformation in recirculating aquaculture systems along with subsequent plant-production processes[1,2]. At the municipal interface, system-level analysis has more often focused on nutrient recovery supply–demand assessment rather than dynamic treatment–cultivation integration[3]. Our project aims to develop a framework that enables system-level simulation of nutrient dynamics across the wastewater–hydroponic interface.

5.1 Mechanistic Modeling of Nitrogen Transformation in Activated Sludge Systems

The activated sludge process is the most widely applied biological wastewater treatment method worldwide, in which bacterial biomass (activated sludge) is responsible for removing pollutants such as excessive nitrogen from wastewater[4]. Various activated sludge configurations have been developed and implemented in wastewater treatment plants, and their corresponding mathematical modeling and simulation have long been essential tools for understanding and optimizing biological nitrogen removal under different operating conditions[5,6]. The ASM1 was the first standardized activated sludge model, followed by later extensions including ASM2, ASM2d, and ASM3[7]. Other widely used models such as Barker & Dold model[8] and the metabolic model developed at the Delft University of Technology (TUDP model)[9] have been published for researchers and wastewater treatment plant operators. Different ASM-based models vary in their targeted substrates and complexity, reflected in the number of processes, state variables, and parameters included[10]. However, all of these models simulate key nitrogen removal processes using a set of state variables and reaction equations based on mass-balance relationships and microbial kinetics[5]. For example, ASM1 describes the dynamic behavior of soluble and particulate substrates using mass-balance differential equations that account for influent flow, reactor volume, and biological reaction rates[6]:

$$\frac{dS_{liq,i}}{dt} = \frac{Q}{V_{liq}} \cdot (S_{in,i} - S_{liq,i}) + \sum_{j=1}^{19} \rho_j v_{i,j} \tag{21}$$

$$\frac{dX_{liq,i}}{dt} = \frac{Q}{V_{liq}} \cdot (X_{in,i} - X_{liq,i}) + \sum_{j=1}^{19} \rho_j v_{i,j} \tag{22}$$

Where: $S_{liq,i}$ is the concentration of each soluble state variable, $X_{liq,i}$ is the concentration of each particulate and biomass state variable, V_{liq} is the liquid reactor volume, Q is the flow in and out of the reactor, $S_{in,i}$ is the input concentration of soluble components, $X_{in,i}$ is the concentration of particulate and biomass components, $\sum_{j=1}^{19} \rho_j v_{i,j}$ is the sum of the specific kinetic rates ρ_j for process j multiplied by the stoichiometric coefficients $v_{i,j}$.

This project will be based on the existing ASM-based mechanistic modeling framework, but the simulation in this project will be designed to achieve controlled nitrogen transformation while retaining a portion of bioavailable nitrogen in the effluent for agricultural reuse. To achieve this goal, selected state variables and kinetic parameters from traditional ASM models will be adjusted to represent partial nitrogen removal rather than full regulatory compliance. The resulting model will remain mechanistic but tailored to nutrient reuse.

5.2 Mechanistic Modeling and Experimental Studies of Hydroponic Cultivation and Reclaimed Wastewater Reuse

Mechanistic simulation of hydroponic plant systems has commonly focused on biomass dynamics together with nutrient uptake and accumulation, particularly nitrogen uptake and tissue nitrate accumulation[11,12]. These studies typically use state-variable ODE frameworks driven by environmental forcing and nutrient availability[11,12]. Representative examples include Michaelis–Menten uptake-based hydroponic dosing simulations validated on Brassica crops[13], as well as soilless greenhouse models for tomato, which update multi-ion solution composition using the uptake-concentration mass-balance approach for scenario and control analysis[14]. Similar kinetic simulation frameworks have also been applied to reclaimed-wastewater lettuce cultivation systems, in which saturating kinetics are used to represent nitrogen uptake and biomass productivity under variable nutrient supply[15].

Experimental studies of reclaimed wastewater nutrient reuse and pilot-scale operations have been more prevalent than integrated mechanistic simulations of wastewater–hydroponic systems. Reclaimed-water hydroponics has more often been investigated through integrated treatment–cultivation pilot systems that evaluate plant growth responses to treated municipal wastewater nutrient levels and typically do not extend to coupled mechanistic simulation[16].

Multi-nutrient growth limitation is commonly represented by Monod-type functions $f_i(S_i) = \frac{S_i}{K_{s,i}+S_i}$ combined across nutrients (e.g., multiplicative co-limitation $\mu = \mu_{max} \prod_i f_i$ or minimum/Liebig limitation), providing a standard starting point for N/P/K-resolved growth simulation[17]. Nutrient uptake and translocation are frequently represented with Michaelis–Menten influx $J_i = J_{max,i} \frac{C_i}{K_{m,i}+C_i}$ and embedded in solution mass balances; this approach has been implemented in hydroponic system simulators using multi-ion uptake with transpiration coupling[13] and in vertical lettuce nutrient film technology modeling where Michaelis–Menten uptake is paired with Monod growth to estimate uptake and biomass responses under nutrient variability[15]. Recent lettuce kinetic studies have extended the Monod multi-nutrient core by introducing Dynamic-μ (maturity-associated decline) with explicit N/P/K limitation, yielding a compact, simulation-ready cultivation kernel coupled to biomass via $\frac{dDM}{dt} = \mu$[18,19]. Complementary mechanistic frameworks for soilless culture formalize the coupling among nutrient and water uptake and root-zone transport processes[20], while controlled-environment lettuce modeling has further combined physics-based transpiration (Penman–Monteith) with structured nutrient-uptake submodels under closed-loop fertigation[21].

Together, these studies provide a foundation of process relationships and governing equations that can be adapted and integrated into physics-driven simulations of wastewater–hydroponic systems. Building on these established mechanistic formulations, the proposed framework seeks to simulate how wastewater-derived nutrient profiles influence hydroponic nutrient uptake, biomass dynamics, and overall system performance across the wastewater–agriculture interface.

6 Current Progress and Initial Modeling Results

6.1 Nitrogen Transformation Simulation and Initial Results

The model includes five state variables: ammonium nitrogen (S_{NH4}), nitrite nitrogen (S_{NO2}), nitrate nitrogen (S_{NO3}), AOB biomass (X_{AOB}), and NOB biomass (X_{NOB}). The external inputs to the system include influent nitrogen concentrations, dissolved oxygen concentration, temperature, pH, and hydraulic retention time. For the base-case simulation, the influent ammonium concentration was set to 40 mg N/L, while influent nitrite and nitrate concentrations were set to 0 mg N/L. The kinetic and stoichiometric parameters used in the model were selected from literature-based typical values for nitrifying systems and adjusted to produce a reasonable demonstration of reactor behavior under stable aerobic operation. Hydraulic conditions were represented using the reactor volume-to-flow ratio (V/Q) and was set to 10 days, which determines the hydraulic retention time and washout intensity. Initial AOB and NOB biomass

concentrations (200 mg/L and 100 mg/L, respectively) were prescribed to represent a reactor that already contains an active nitrifying microbial community at startup.

Listing 1 details the core Python implementation of the numerical simulation, which is structured around two primary functions. The system dynamics function evaluates the five coupled ordinary differential equations governing the reactor. It calculates the microbial growth rates using dual-substrate Monod kinetics, incorporating both nitrogen availability and dissolved oxygen limitations. These kinetic rates are subsequently applied to the mass balance equations to determine the rate of change for soluble nitrogen species and particulate biomass fractions. To simulate the temporal evolution of these state variables, a secondary function applies a fourth-order Runge-Kutta integration scheme, ensuring numerical stability when resolving the non-linear biological growth rates.

Listing 1: Core System Dynamics and RK4 Integration Scheme

```
def system_dynamics(y, t, p):
    S_NH4, S_NO2, S_NO3, X_AOB, X_NOB = y

    # Monod growth kinetics with DO limitation
    mu_AOB = p['mu_AOB_max'] * (S_NH4 / (p['K_NH4'] + S_NH4)) * \
              (p['SO2'] / (p['K_O2_AOB'] + p['SO2'])) * X_AOB
    mu_NOB = p['mu_NOB_max'] * (S_NO2 / (p['K_NO2'] + S_NO2)) * \
              (p['SO2'] / (p['K_O2_NOB'] + p['SO2'])) * X_NOB

    S_NH4_in = 40.0  # Influent ammonia concentration (mg/L)

    # Mass balances for Nitrogen species
    dNH4 = (1/p['V_Q'])*(S_NH4_in - S_NH4) - (1/p['Y_AOB'])*mu_AOB
    dNO2 = (1/p['V_Q'])*(0 - S_NO2) + (1/p['Y_AOB'])*mu_AOB - (1/p['Y_NOB'])*mu_NOB
    dNO3 = (1/p['V_Q'])*(0 - S_NO3) + (1/p['Y_NOB'])*mu_NOB

    # Mass balances for Microbial Biomass
    dX_AOB = mu_AOB - p['b_AOB']*X_AOB - (1/p['V_Q'])*X_AOB
    dX_NOB = mu_NOB - p['b_NOB']*X_NOB - (1/p['V_Q'])*X_NOB

    return np.array([dNH4, dNO2, dNO3, dX_AOB, dX_NOB])

def rk4_step(y, t, dt, p):
    k1 = system_dynamics(y, t, p)
    k2 = system_dynamics(y + dt/2 * k1, t + dt/2, p)
    k3 = system_dynamics(y + dt/2 * k2, t + dt/2, p)
    k4 = system_dynamics(y + dt * k3, t + dt, p)
    return y + dt/6 * (k1 + 2*k2 + 2*k3 + k4)
```

Figure 1 illustrates the overall performance of the system over a 5-day period.

The simulation showed that nitrification in the reactor was highly stable in terms of nitrogen conversion. Ammonium was removed efficiently, nitrite accumulation remained limited, and nitrate was the dominant end product over time. These trends indicate that both nitrifying steps were successfully maintained under the selected dissolved oxygen, temperature, and hydraulic conditions. Although AOB and NOB biomass decreased initially, this does not contradict the stable nitrification performance. Because the biomass equations include decay and hydraulic washout, the initially assigned biomass concentrations were higher than the long-term biomass level required by the system under these conditions. As the reactor operated, the microbial populations adjusted downward and gradually approached a more stable level. Thus, the observed biomass decline is better interpreted as a transition toward a new equilibrium rather than as biological instability. In this model, the sustained conversion of ammonium to nitrate demonstrates that the nitrification process remained stable even though biomass concentrations evolved over time, suggesting a high process stability in this system.

In addition to stable treatment performance, the simulated effluent nitrogen composition

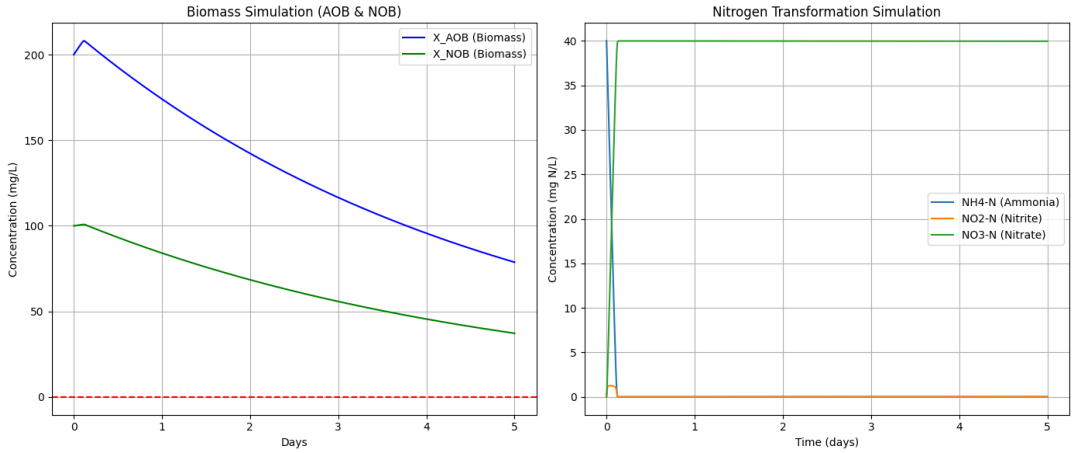

Figure 1: Preliminary simulation of coupled biomass growth and nitrogen transformation

also satisfied the original design objective of this project. Most of the influent ammonium was converted into nitrate, while ammonium and nitrite remained at relatively low concentrations in the effluent. This nitrogen profile is desirable for downstream hydroponic vegetable production, since nitrate is generally a more suitable nitrogen source for plant uptake, whereas high ammonium concentrations may be less favorable for crop growth. Therefore, the modeled reactor not only achieved stable nitrification, but also generated an effluent composition that is suitable for coupling with the downstream vegetable growth model. These results support the overall design concept of integrating wastewater treatment and hydroponic crop production through controlled biological nitrogen transformation.

6.2 Hydroponic Plant Growth Simulation and Initial Results

6.2.1 Model implementation

Building on previous studies, the present plant growth module is formulated as a general mechanistic crop-growth simulator using an ODE framework in which dry-mass growth is governed by nutrient availability and related forcing through specific growth-rate formulations and a biomass governing equation. In the current workflow, dry mass (DM) is used as the primary biomass state variable, which is the simulated target variable of the models. To represent alternative ways of describing nutrient-limited plant growth, the current implementation includes five candidate formulations for the specific growth rate, μ: a nitrogen-focused Dynamic-μ model, a category-aware RMND-μ model, a shared SMND-μ model, a Liebig minimum Monod model, and a multiplicative Monod model. The Liebig minimum Monod model (Eq. 14) and the multiplicative Monod model (Eq. 15) are the most straightforward Monod-type growth models in the present module. Since they require only $\mu_{\max}$ and nutrient half-saturation coefficients, they can be used directly as scenario models when nutrient concentrations are known and parameter values are assigned from literature or prior assumptions. In contrast, the Dynamic-μ model (Eq. 16), the RMND-μ model (Eq. 17), and the SMND-μ model (Eq. 18) introduce additional time-dependent or category-dependent parameters and are therefore more data-dependent. Thus, while all five formulations use nutrient concentrations to compute μ, the simpler Monod-based models are more practical in no-data situations, whereas the Dynamic-μ, RMND-μ, and SMND-μ formulations are most appropriate when experimental biomass data are available for calibration.[17,18,19]

Listing 2 presents the core implementation of the plant growth simulation. The code first defines nutrient-limitation functions and representative growth-rate expressions. These growth-rate terms are then incorporated into the biomass governing equation in Eq. 12. To simulate the temporal evolution of dry mass, the system applies a fourth-order Runge–Kutta integration scheme over the cultivation period.

Listing 2: Core growth-rate formulations and RK4-based dry-mass simulation

```
# Monod saturation function for one nutrient
def frac_monod(s, ks):
    s = max(float(s), 1e-12)
    ks = max(float(ks), 1e-12)
    return s / (ks + s)

# Dynamic- model with nitrogen limitation and maturity-related decline
def calc_mu_dynamic(t, forcing, params, category):
    N = forcing_N_at_time(t, forcing)
    mu = params["mu_max"] * frac_monod(N, params["Ks"])
    mu += params["m"] * max(0.0, t - params["delay"])
    return mu

# Liebig minimum Monod model controlled by the most limiting nutrient
def calc_mu_liebig(t, forcing, params, category):
    N = forcing_N_at_time(t, forcing)
    P = forcing_P_at_time(t, forcing)
    K = forcing_K_at_time(t, forcing)
    fmin = min(
        frac_monod(N, params["KsN"]),
        frac_monod(P, params["KsP"]),
        frac_monod(K, params["KsK"]),)
    return params["mu_max"] * fmin

# Multiplicative Monod model with simultaneous nutrient co-limitation
def calc_mu_multiplicative(t, forcing, params, category):
    N = forcing_N_at_time(t, forcing)
    P = forcing_P_at_time(t, forcing)
    K = forcing_K_at_time(t, forcing)
    mu = params["mu_max"]
    mu *= frac_monod(N, params["KsN"])
    mu *= frac_monod(P, params["KsP"])
    mu *= frac_monod(K, params["KsK"])
    return mu

# Dry-mass balance: dDM/dt =  * DM
def biomass_ode(y, t, forcing, params, category, model_name):
    B = y[0]
    if model_name == "dynamic":
        mu = calc_mu_dynamic(t, forcing, params, category)
    elif model_name == "liebig":
        mu = calc_mu_liebig(t, forcing, params, category)
    elif model_name == "multiplicative":
        mu = calc_mu_multiplicative(t, forcing, params, category)
    else:
        raise ValueError(f"Unknown model_name: {model_name}")
    dBdt = mu * B
    return np.array([dBdt])

# One RK4 integration step for the biomass state
def rk4_step_scalar(ode_fun, y, t, dt, *args):
    k1 = ode_fun(y, t, *args)
    k2 = ode_fun(y + dt / 2 * k1, t + dt / 2, *args)
    k3 = ode_fun(y + dt / 2 * k2, t + dt / 2, *args)
    k4 = ode_fun(y + dt * k3, t + dt, *args)
    return y + dt / 6 * (k1 + 2 * k2 + 2 * k3 + k4)

# Simulate dry-mass trajectories over the cultivation period
def simulate_biomass(model_name, forcing, params, category, B0, days_obs):
```

```
    t_dense = np.linspace(0, X_MAX, int(X_MAX * ODE_STEPS_PER_DAY) + 1)
    dt = t_dense[1] - t_dense[0]
    y = np.array([B0], dtype=float)
    sol = []
    for ti in t_dense:
        sol.append(y.copy())
        y = rk4_step_scalar(
            biomass_ode, y, ti, dt,
            forcing, params, category, model_name)
    sol = np.array(sol)
    B_dense = sol[:, 0]
    B_fit = np.interp(days_obs, t_dense, B_dense)
    return B_fit
```

6.2.2 Experimental data usage: parameter fitting and model evaluation

The current implementation and initial testing use lettuce data collected from a vertical nutrient film technique (NFT) system operated with recovered nutrients from reclaimed wastewater under nutrient-limited conditions. These data are used not only for model evaluation, but also for calibration of the growth-model parameters. Therefore, the dataset serves not merely as a case-specific validation source, but as the experimental basis for establishing calibrated parameter sets intended to represent lettuce growth behavior under the modeled reclaimed-water hydroponic conditions. These calibrated parameter sets can subsequently be carried forward into the integrated framework, where nutrient concentrations predicted by the upstream wastewater-treatment module are supplied as inputs to the downstream plant cultivation module.

The original data used in this study are openly available in the USDA Ag Data Commons repository as *Dataset for the Hydroponic Cultivation of Bibb Lettuce in NPK-Limited Conditions*, DOI: `https://doi.org/10.15482/USDA.ADC/28801286.v1`, accessed on 22 August 2025. For the present workflow, the fitting data are taken from the `ModSummary` sheet, whereas the plotting and validation data are taken from the `ModPlant` sheet. Treatment-level nutrient values for N, P, and K are extracted from the dataset and used as forcing inputs to the cultivation simulations. Although the full downstream dataset and analysis include N-, P-, K-limited, and MSS treatments (nutrient-sufficient control), the results and discussion presented in this report focus primarily on the N-limited cases in order to align with the current upstream nitrogen-transformation analysis.

The workflow first loads the lettuce NFT dry-mass data, removes excluded rows such as invalid growth stages or screening rows, fills nutrient inputs using sample-period or lifetime-average values, assigns treatment categories, and then aggregates observations by treatment and growth day. Constant treatment-level N, P, and K values are subsequently passed into the simulation as forcing inputs. Listing 3 shows the key treatment-forcing implementation.

Listing 3: Experimental data loading and treatment-level forcing construction

```
# Construct constant treatment-level nutrient forcing inputs
def build_treatment_forcing(sub_fit):
    N0 = float(sub_fit["N"].iloc[0])
    P0 = float(sub_fit["P"].iloc[0])
    K0 = float(sub_fit["K"].iloc[0])
    forcing = {
        "t": np.array([0.0, X_MAX], dtype=float),
        "N": np.array([N0, N0], dtype=float),
        "P": np.array([P0, P0], dtype=float),
        "K": np.array([K0, K0], dtype=float),
    }
    return forcing
```

Parameter fitting is currently performed using dry mass measurements. The main fitting-related functions are `fit_dynamic_dm(...)`, `fit_rmnd_dm(...)`, `fit_smnd_dm(...)`, `fit_liebig_dm(...)`,

and `fit_multiplicative_dm(...)`. For each growth formulation, the corresponding parameter vector is estimated by minimizing the sum of squared differences between observed and simulated dry mass according to Eq. 19 and Eq. 20. This process yields a calibrated growth-model parameter set for each formulation. Listing 4 shows the Dynamic-μ fitting routine as a representative example. The same overall optimization structure is used for the other formulations, with the parameter vector adjusted according to the selected growth-rate model.

Listing 4: Representative dry-mass parameter fitting routine for the Dynamic-μ model

```
# Fit Dynamic- parameters by minimizing dry-mass prediction error
def fit_dynamic_dm(fit_df):
    init = MODEL_INIT["dynamic"]

    def objective(theta):
        mu_max, Ks, m, delay = theta
        params = {
            "mu_max": mu_max,
            "Ks": Ks,
            "m": m,
            "delay": delay,}

        total = 0.0
        for treatment in sorted(fit_df["Treatment"].unique()):
            # Extract one treatment time series for fitting
            sub_fit = fit_df[fit_df["Treatment"] == treatment].sort_values("Growth Day
    ")
            category = sub_fit["Category"].iloc[0]
            forcing = build_treatment_forcing(sub_fit)

            # Use observed dry mass at sampled growth days
            days_obs = sub_fit["Growth Day"].values.astype(float)
            y_dm_obs = sub_fit["Y_DM"].values.astype(float)
            B0_dm = float(y_dm_obs[0])

            # Simulate dry-mass trajectory for the current parameter set
            y_dm_pred = simulate_biomass(
                "dynamic", forcing, params, category, B0_dm, days_obs
            )

            # Accumulate squared prediction error
            total += np.sum((y_dm_obs - y_dm_pred) ** 2)
        return total

    # Optimize parameters subject to bounds
    x0 = np.array([init["mu_max"], init["Ks"], init["m"], init["delay"]], dtype=float)
    bounds = [(1e-6, 2.0), (1e-6, 500.0), (-0.2, 0.0), (0.0, 40.0)]
    res = minimize(objective, x0=x0, bounds=bounds, method="L-BFGS-B")
```

6.2.3 Validation and visualization using data

After calibration, the module uses experimental dry-mass data again for validation and visualization. The main functions used for this stage are `load_plot_data_dm(...)`, `evaluate_one_model(...)`, and `plot_category_panels_overlay(...)`. For each model, the fitted parameters are used to simulate treatment-level dry-mass trajectories, and the predicted dry mass at the measured growth days is compared with the observed dry-mass data. Standard regression metrics, including R^2, MAE, and RMSE, are then calculated.

For visualization, all candidate growth formulations can be overlaid within the same treatment subplot so that differences among model formulations can be inspected directly for each

retained treatment. Listing 5 shows the key evaluation workflow used to generate treatment-level prediction results.

Listing 5: Representative dry-mass evaluation workflow

```
def evaluate_one_model(plot_df_dm, model_name, params):
    # Evaluate one fitted model by treatment
    results = {}
    treatment_rows_dm = []

    for treatment, sub_plot in plot_df_dm.groupby("Treatment"):
        # Sort observations by growth day
        sub_plot = sub_plot.sort_values("Growth Day").reset_index(drop=True)
        sub_fit = fit_df_dm[fit_df_dm["Treatment"] == treatment].sort_values("Growth
    Day").reset_index(drop=True)

        # Get treatment category, forcing inputs, and observed dry-mass data
        category = sub_fit["Category"].iloc[0]
        forcing = build_treatment_forcing(sub_fit)
        days_obs = sub_fit["Growth Day"].values.astype(float)
        y_dm_obs = sub_fit["Y_DM"].values.astype(float)
        B0_dm = float(y_dm_obs[0])

        # Simulate treatment-level dry-mass trajectory
        t_grid, y_dm_plot, y_dm_fit = simulate_biomass(
            model_name, forcing, params, category, B0_dm, days_obs)

        # Calculate goodness-of-fit metrics
        met_dm = metrics_dict(y_dm_obs, y_dm_fit)

        treatment_rows_dm.append({
            "Model": MODEL_DISPLAY[model_name],
            "Category": display_category_label(category),
            "Treatment": display_treatment_label(treatment),
            "R2": met_dm["R2"],
            "MAE": met_dm["MAE"],
            "RMSE": met_dm["RMSE"],})
```

At this stage, the module successfully produces calibrated parameter values, dry-mass simulations, evaluation metrics, and comparative plots for the candidate growth formulations. The present nutrient forcing is based on treatment-level experimental inputs rather than fully dynamic upstream-downstream coupling, which will be updated in the next step.

6.2.4 Validation and visualization using data

After calibration, the fitted parameter sets were used to simulate treatment-level dry-mass trajectories and to compare predicted values against observed dry-mass measurements at the sampled growth days. Model performance was evaluated using standard regression metrics, including R^2, MAE, and RMSE. Comparative plots were then generated by overlaying all candidate growth formulations within the same treatment panel so that differences in predictive behavior could be assessed directly across retained treatments.

Table 1 summarizes the aggregate dry-mass prediction performance of the five candidate growth formulations. At the current stage, the evaluation results indicate clear differences in dry-mass predictive performance among the five candidate formulations. Among the tested models, RMND-μ and SMND-μ showed the best overall performance and yielded identical aggregate metrics in the current implementation, followed by the Liebig minimum Monod and multiplicative Monod models. The Dynamic-μ model showed the weakest overall performance across the retained treatments.

Although aggregate metrics provide an overall ranking, treatment-level behavior remained

Table 1: Overall dry-mass prediction performance of the five candidate growth formulations.

Model	R^2	MAE	RMSE
Dynamic-μ	0.6939	0.6199	1.3300
RMND-μ	0.8308	0.4706	0.9889
SMND-μ	0.8308	0.4706	0.9889
Liebig minimum Monod	0.7827	0.5689	1.1207
Multiplicative Monod	0.7526	0.5960	1.1958

heterogeneous. Figure 2 presents the treatment-level dry-mass trajectories for the N-limited cases, comparing the observed data with the fitted simulations. The Dynamic-μ model followed the observed trajectories reasonably well for the 25 and 66 mg L^{-1} N treatments, but it underpredicted biomass in several higher-concentration cases and did not provide the strongest overall agreement across the full set of N-limited treatments. The RMND-μ and SMND-μ formulations generally provided closer agreement for several intermediate and high-growth N-limited cases, although both remained weak in the most extreme conditions, including the 5, 11, and 264 mg L^{-1} N treatments. The Liebig minimum Monod and multiplicative Monod models captured the general increase in biomass with nitrogen availability, but they tended to produce steeper late-stage growth in several higher-nitrogen treatments.

These results suggest that, in the current downstream implementation, the multi-nutrient Dynamic-μ-based formulations provide the strongest overall representation of dry-mass growth under reclaimed-wastewater nutrient-limited conditions. At the same time, the fact that RMND-μ and SMND-μ currently produce identical aggregate evaluation results indicates that the distinction between these two formulations should be interpreted cautiously at this stage and examined further in the present code and parameterization workflow. Meanwhile, the Liebig minimum Monod model provided a stronger baseline performance than the multiplicative Monod model, suggesting that the minimum-limitation assumption may better represent the present dataset than fully multiplicative co-limitation. Overall, the plant growth module now successfully generates calibrated parameter values, treatment-level dry-mass simulations, evaluation metrics, and comparative visualizations for all five growth formulations. These results should still be interpreted as downstream-module validation results rather than fully integrated wastewater–hydroponic simulation results, since the current nutrient forcing remains based on treatment-level experimental inputs rather than dynamically simulated upstream outputs.

7 Work Distribution

So far, we have already developed two separate subsystem models: a wastewater nitrification model and a hydroponic plant growth model. Both model has been implemented and tested independently, and are able to produce reasonable simulation results under simplified assumptions. In addition, the plant growth simulation module has already been extended to include parameter calibration using plant growth experimental data, as well as model validation and visualization based on observed dry-mass trajectories.

Building on this progress, the next stage of this project will focus on three main tasks: model coupling, integrated simulation using upstream outputs, and further improvement of visualization and result interpretation. The work will be divided among four team members to ensure efficient progress.

Zijun Meng will focus on validating and improving the wastewater treatment model. This includes collecting nitrogen concentration data from literature or public datasets, comparing it with model outputs, and adjusting parameters to better match real conditions.

Yuming Sun will focus on refining the downstream hydroponic plant growth model and integrating it with the upstream wastewater-treatment simulation. Specifically, she will use the simulated recovered nutrient outputs from the wastewater module as input nutrient concentrations for the downstream plant growth model and evaluate how upstream nitrogen transformation

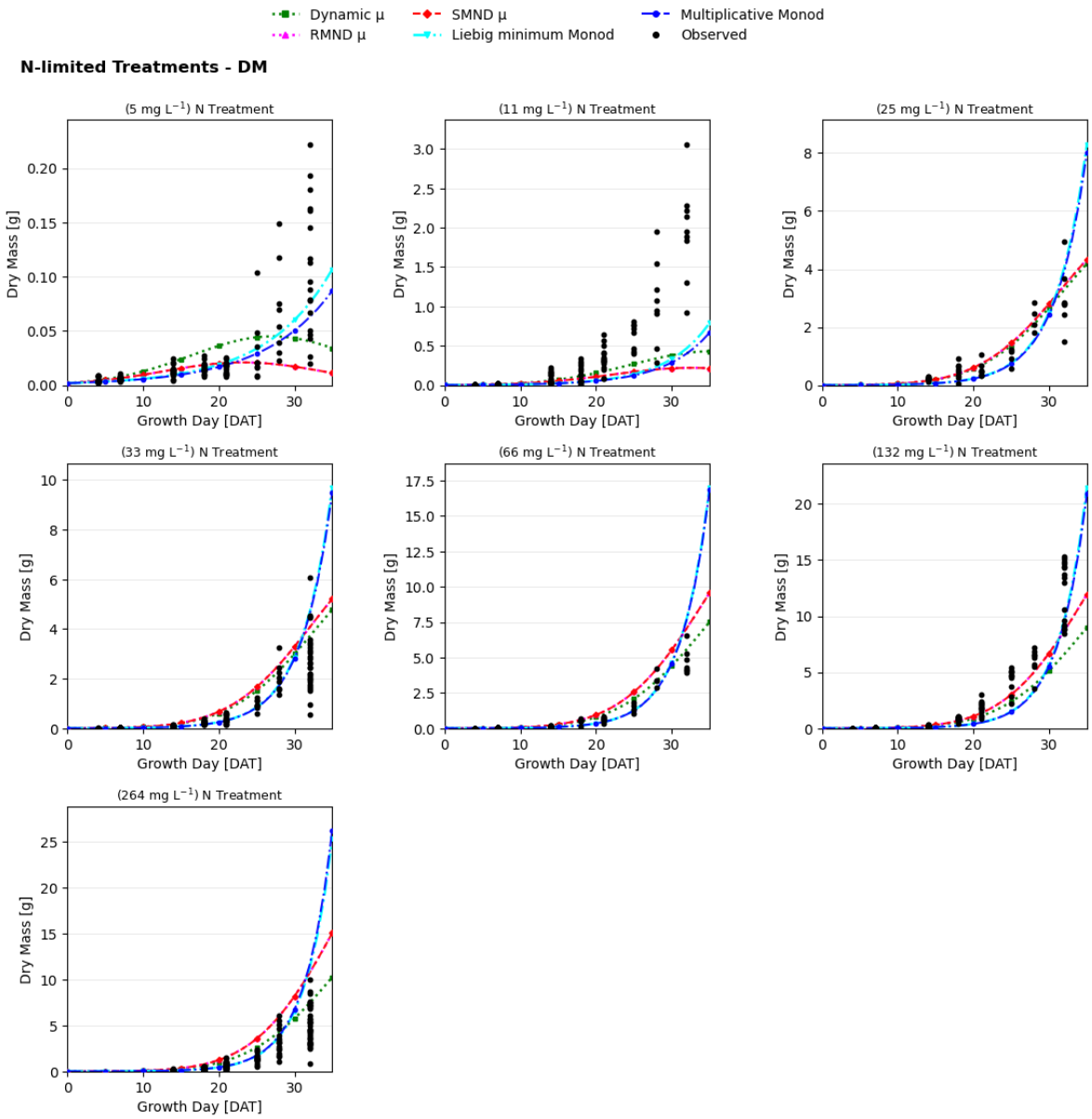

Figure 2: N-limited Plant Growth: observed and simulated dry-mass trajectories under N-limited treatments using the Dynamic-μ and Liebig minimum Monod models.

affects downstream biomass growth under nutrient-limited conditions.

Ruixin Liu will be responsible for coupling the two models and building the integrated system. This includes connecting the output of the wastewater model to the input of the hydroponic model, and ensuring the simulation runs in a stable and consistent way.

Jiasheng Cao will focus on result analysis and visualization. This includes improving the current plots and developing more intuitive ways to show the system behavior, such as multi-variable plots and interactive visualizations, to better illustrate the relationship between nitrogen transformation and plant growth.

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